

MERJEM ALIĆ

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OBJECTIVE

Software Engineering student with a strong foundation in algorithms, systems-level development, and problem solving, seeking an internship to build and ship production software.

EDUCATION

Bachelor of Information Technology, International Burch University
Software Engineering Track

Expected 2027

Relevant Coursework: Parallel Programming, Object-Oriented Programming, Data Structures and Algorithms, Operating Systems, Database Systems, Optimization Algorithms.

SKILLS

Languages	Java, C, C++, Python, JavaScript, PHP, Kotlin, HTML, CSS
Frameworks	Pyomo, FlightPHP, Jetpack Compose, Bootstrap
Databases	SQL, MySQL, MongoDB, Redis, PostgreSQL
Tools	Version Control (Git, GitHub), Claude Code, Maven, Postman

EXPERIENCE

Backend AI Engineering Intern
FlyRank AI

July 2026 - September 2026
Chicago, IL - Remote

Developed backend services with Express, SQLite, Supabase, and Docker, and completed structured AI-fluency training in agent design and workflow evaluation. Shipped an LLM-powered social media content pipeline with a crash-safe scheduler as a final project. Focused on writing clean, well-tested code and building strong fundamentals across the stack.

PROJECTS

FinchOS. Building a custom x86-64 operating system from scratch in C and assembly, working toward a fully functioning OS with a graphical user interface. Implemented bootloader integration, a custom GDT, a full IDT with exception handling, PIC/PIT-driven interrupt handling, a keyboard driver, memory management, and a preemptive round-robin scheduler with context switching.

QEMU MCP. C/C++ Model Context Protocol server for live kernel debugging in QEMU, built to inspect FinchOS. Speaks the GDB remote serial protocol directly over sockets, exposing tools for register/memory inspection, breakpoints, and step execution.

Academiq. Native Android app centralizing course management, assignment tracking, and study scheduling, built with Kotlin, Jetpack Compose, and MVVM. Modular multi-screen UI with reusable components, backed by RoomDB for local storage and Firebase Authentication/Firestore for cloud sync.

Car Dealership Web Application. Full-stack single-page dealership platform using JavaScript, PHP, FlightPHP, and MySQL. Three-tier architecture with RESTful APIs, JWT authentication, full CRUD operations, a PDO-based DAO data layer, and production deployment on DigitalOcean.

Vibe Guard (Reddit Moderation Tool). Asynchronous moderation platform built with TypeScript, Devvit Web SDK, and Hono. Integrates the OpenAI Moderation API for real-time toxicity detection and a Redis-backed state machine for tracking user history and applying tiered enforcement (warnings, deletions, thread lockdowns).

CERTIFICATIONS

- Introduction to Parallel Programming with CUDA - Johns Hopkins University (Coursera)
- Claude Code in Action - Anthropic
- Model Context Protocol: Advanced Topics - Anthropic